01-13B INTAKE-AIR SYSTEM [WL-C, WE-C]

01

INTAKE AIR SYSTEM OUTLINE [WL-C, WE-C]

dcf011300000t05

Features

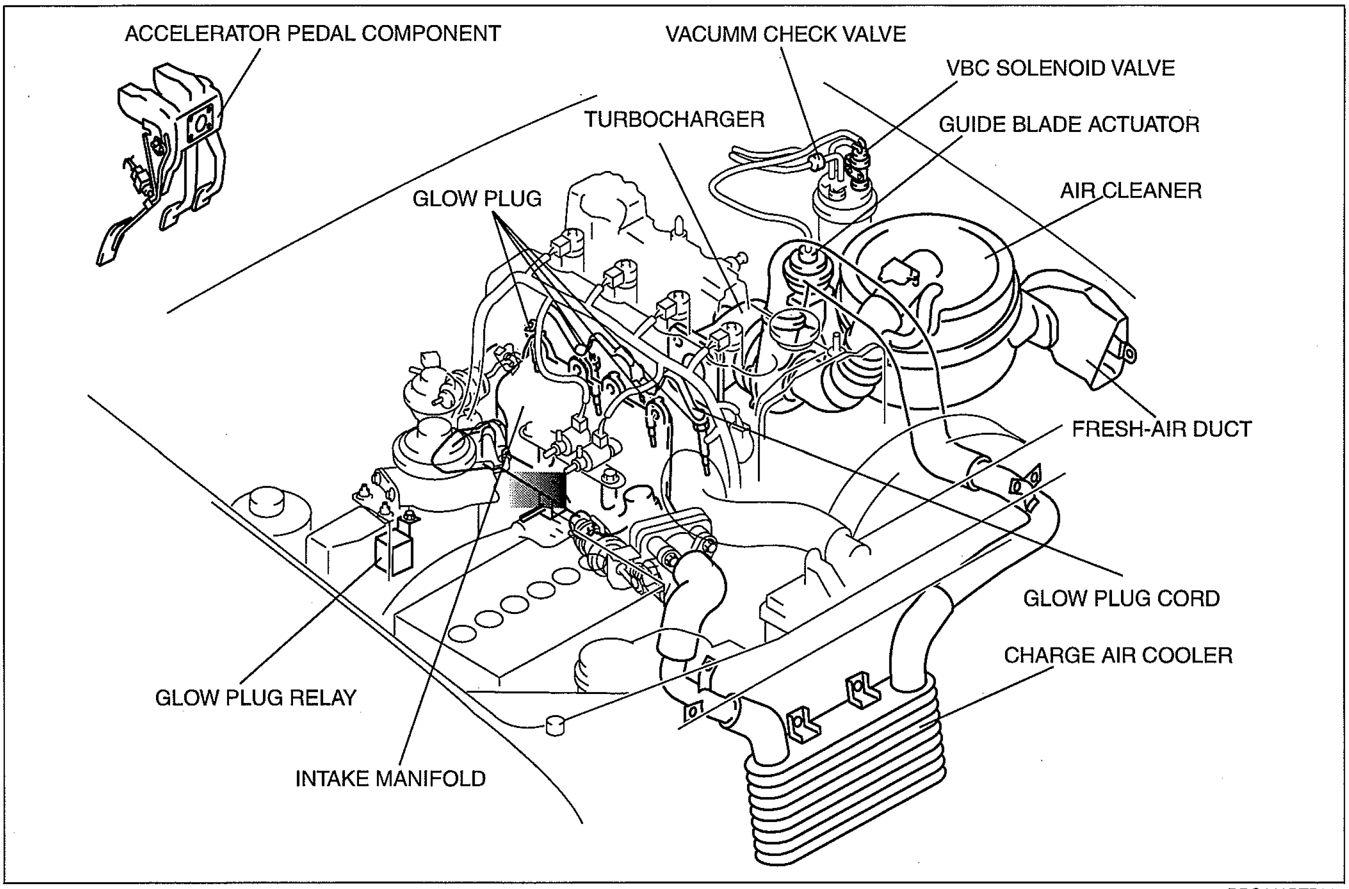
Power efficiency, performance, and fuel economy	• A variable geometry turbocharger has been adopted

Specification

Item	WL-C, WE-C
Turbocharger type	Variable geometry turbocharger
Air cleaner element	Dry type
Glow plug type	Stainless type

INTAKE AIR SYSTEM STRUCTURAL VIEW [WL-C, WE-C]

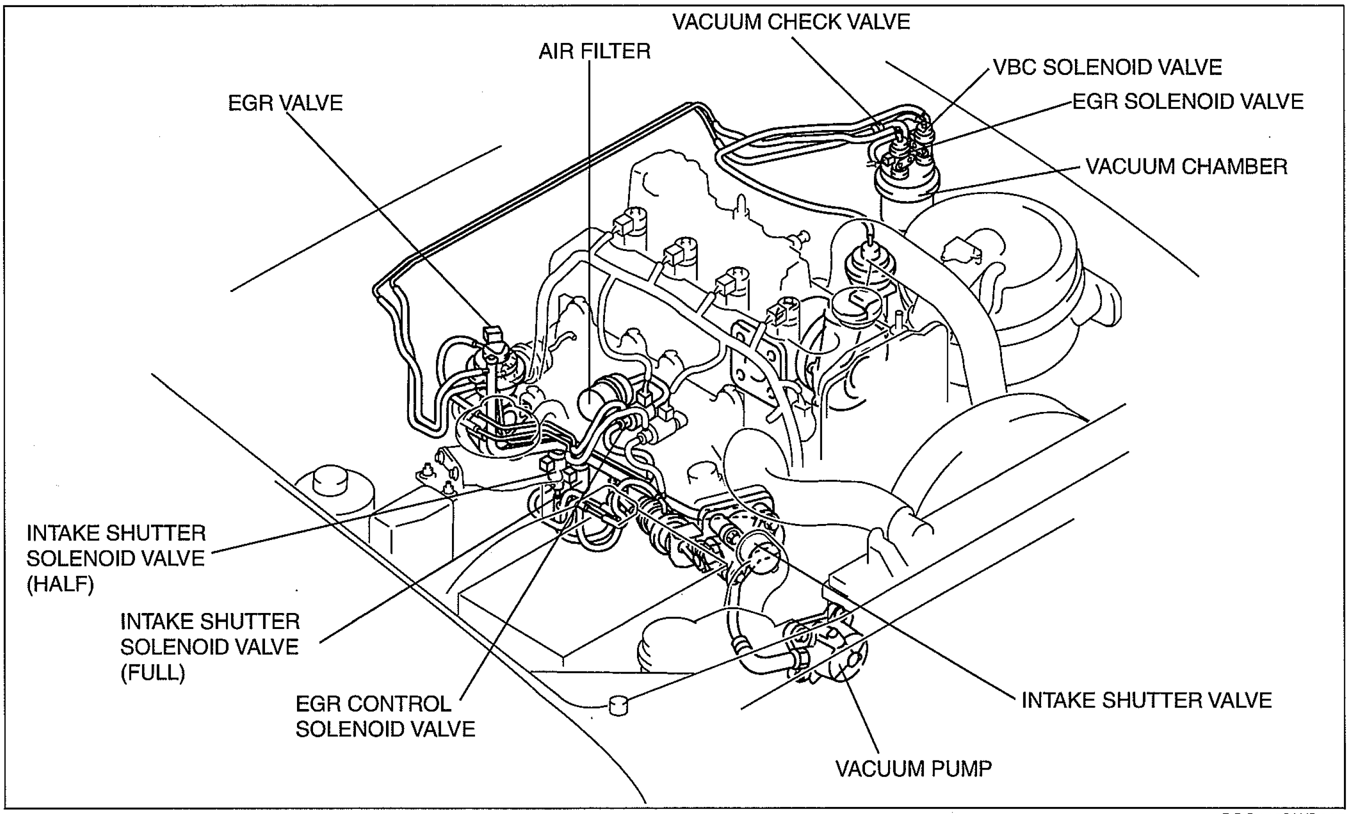
dcf011300000t06



DBG113BTB805

INTAKE AIR SYSTEM HOSE ROUTING DIAGRAM [WL-C, WE-C]

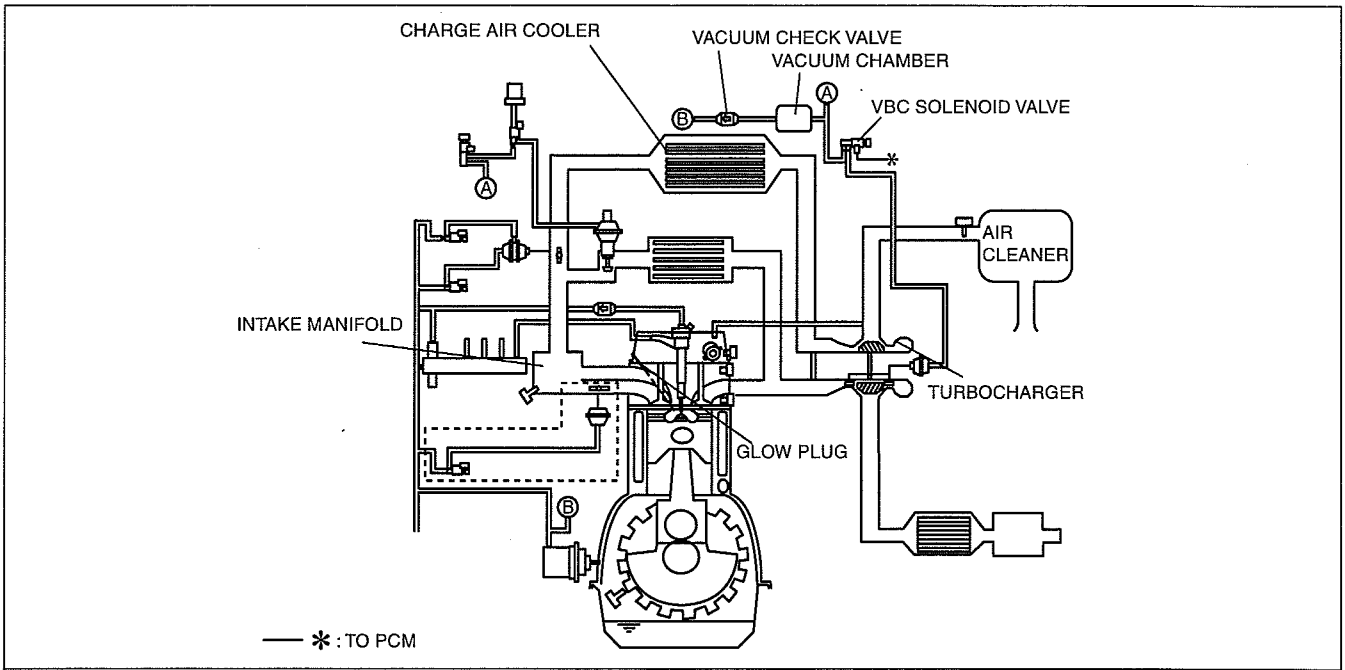
dcf011300000t07



DBG113BWB305

INTAKE AIR SYSTEM DIAGRAM [WL-C, WE-C]

dcf011300000t08

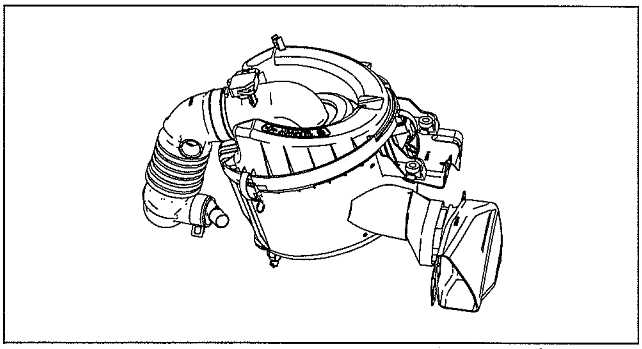


DGB113BTB001

AIR CLEANER CONSTRUCTION [WL-C, WE-C]

dcf011313300t02

- Consists of the air cleaner case, air cleaner element, and fresh-air duct.
- A dry type air cleaner element has been adopted.



DGB113BTB002

VARIABLE BOOST CONTROL (VBC) SOLENOID VALVE FUNCTION [WL-C, WE-C]

dcf011318748t01

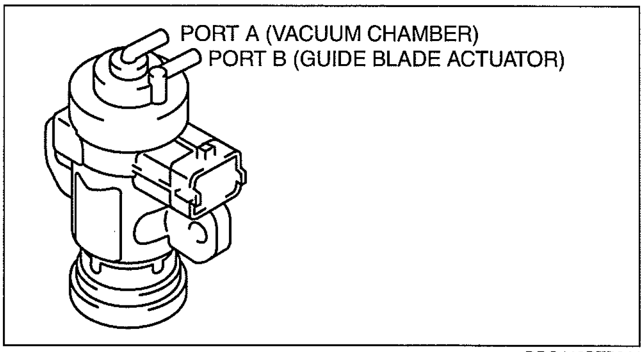
- Switches the vacuum passage between the vacuum chamber and the guide blade actuator.
- Vacuum is applied to the guide blade actuator using the VBC solenoid valve.
- As a result, the variable guide vanes of the turbocharger are adjusted based on the signals from the PCM.

VARIABLE BOOST CONTROL (VBC) SOLENOID VALVE CONSTRUCTION [WL-C, WE-C]

dcf011318748t02

Construction

- Consists of the solenoid coil, spring, and plunger.



DBG113BTB311

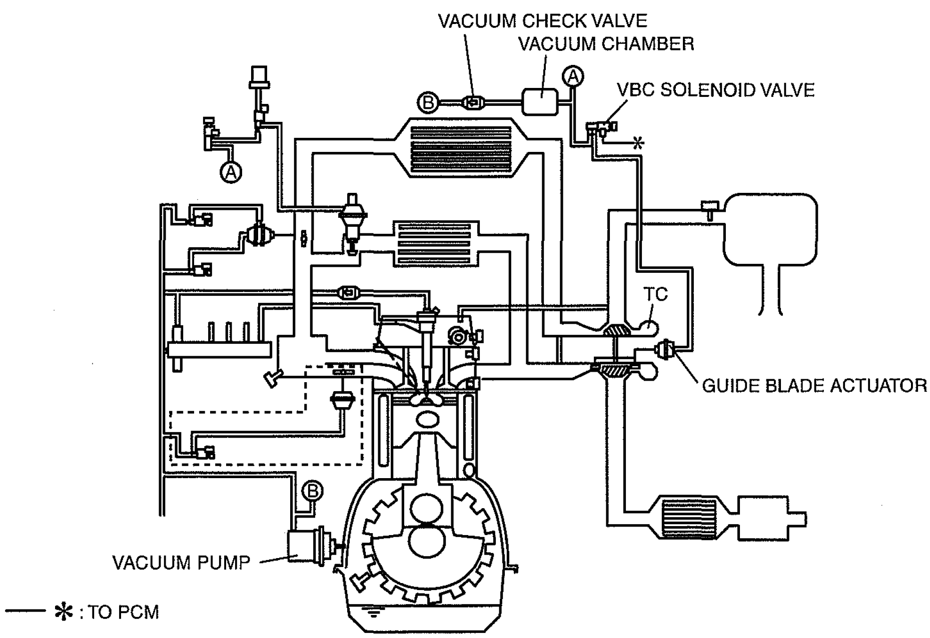
AIR CHARGING SYSTEM OUTLINE [WL-C, WE-C]

dcf011313700t04

- A variable boost (variable geometry) type turbocharger has been adopted. By changing the opening angle of the guide blades and changing the exhaust pressure that acts on the turbine impellers, the optimum boost pressure according and fuel consumption have been improved in every driving range.

AIR CHARGING SYSTEM DIAGRAM [WL-C, WE-C]

dcf011313700t05

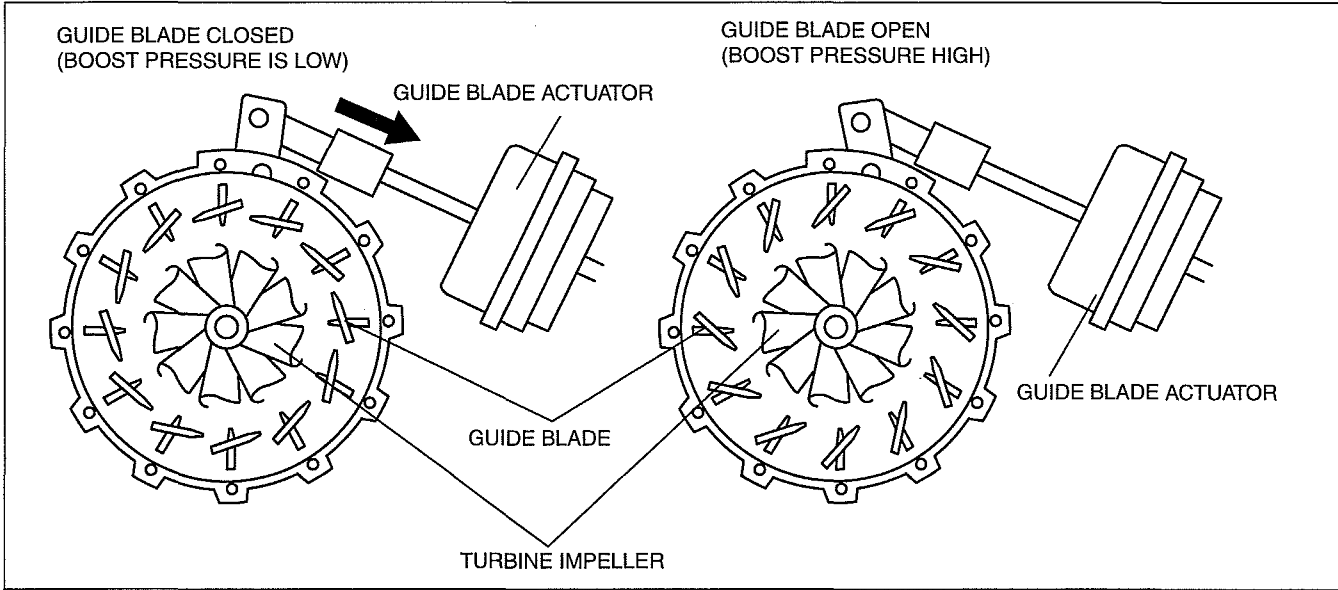


DBG113BTB702

AIR CHARGING SYSTEM OPERATION [WL-C, WE-C]

dcf011313700t06

- When the VBC solenoid valve is operated according to the duty signal from the PCM, the vacuum passage is opened and vacuum is applied to the guide blade actuator of the turbocharger. Due to this, the rod of the guide blade actuator is pulled and the opening angle of the guide blades changes.
- When the duty signal is on for a long time, the vacuum that acts on the guide blade actuator is reduced and the guide blades move in the closing direction.
- When the duty signal is off for a long time, the vacuum that acts on the guide blade actuator is reduced and the guide blades move in the opening direction.
- When the guide blades are moved in the closing direction, the passages that lead the exhaust gas to the turbine impellers become narrow and the flow rate of the exhaust gas increases. Due to this, the exhaust gas pressure is increased and the boost pressure is also increased. On the other hand, when the guide blades are moved in the opening direction, the passages that lead the exhaust gas to the turbine impellers become wide. Due to this, the exhaust gas pressure is reduced and the boost pressure is also reduced.



DBG113BTB802

TURBOCHARGER FUNCTION [WL-C, WE-C]

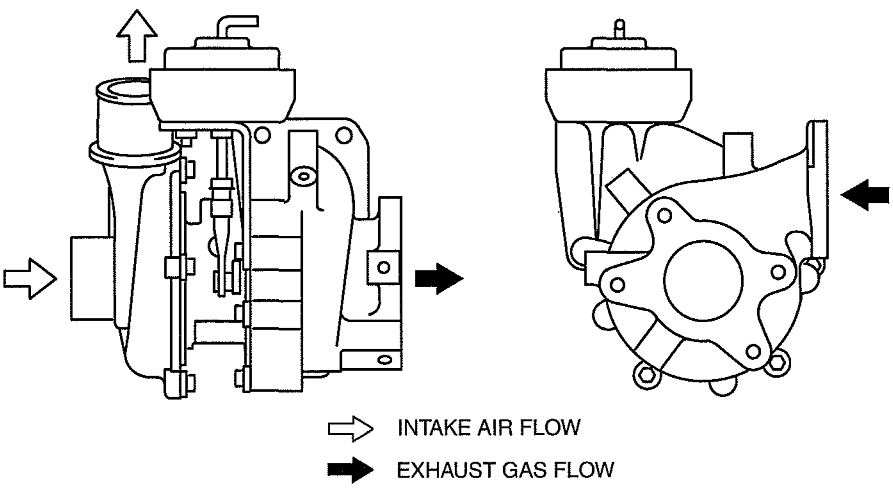
dcf011313700t07

- The turbocharger uses the exhaust gas pressure to compress the intake air.

TURBOCHARGER CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf011313700t08

- A variable geometry turbocharger has been adopted.
- The turbocharger consists of mainly a turbine wheel, a compressor wheel, guide blades, and a guide blade actuator.
- When the exhaust gas flows to the turbine wheel, the turbine wheel and the compressor wheel that is on the same axis as the turbine wheel rotate to compress the intake air.
- The opening angle of the guide blades changes according to the movement of the guide blade actuator rod.

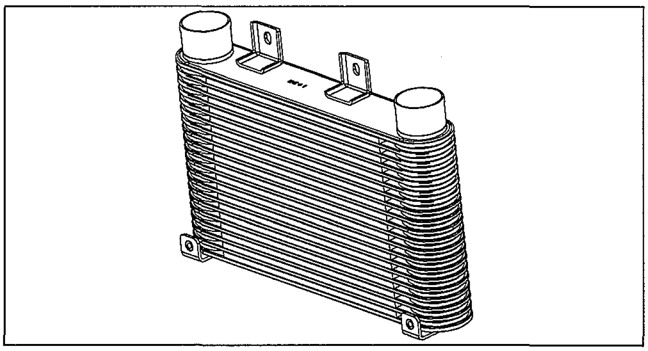


DBG113BTB396

CHARGE AIR COOLER CONSTRUCTION [WL-C, WE-C]

dcf011313560t02

- An aluminium-alloy intercooler has been adopted.
- Compressed and heated intake air is cooled by the intake air charger to increase air density, enhancing the charging efficiency.

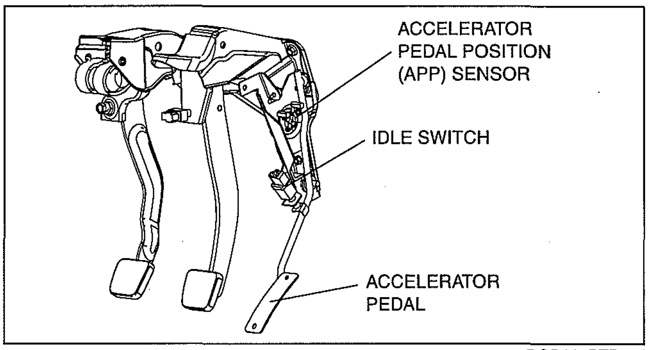


DGB113BTB003

ACCELERATOR PEDAL COMPONENT CONSTRUCTION [WL-C, WE-C]

dcf011341600t02

- Consists of the accelerator pedal and accelerator pedal position (APP) sensor.



DGB113BTB004

GLOW SYSTEM OUTLINE [WL-C, WE-C]

dcf011318601t07

- The glow system warms up the combustion chamber at engine start to improve ignitability.

GLOW SYSTEM CONSTRUCTION [WL-C, WE-C]

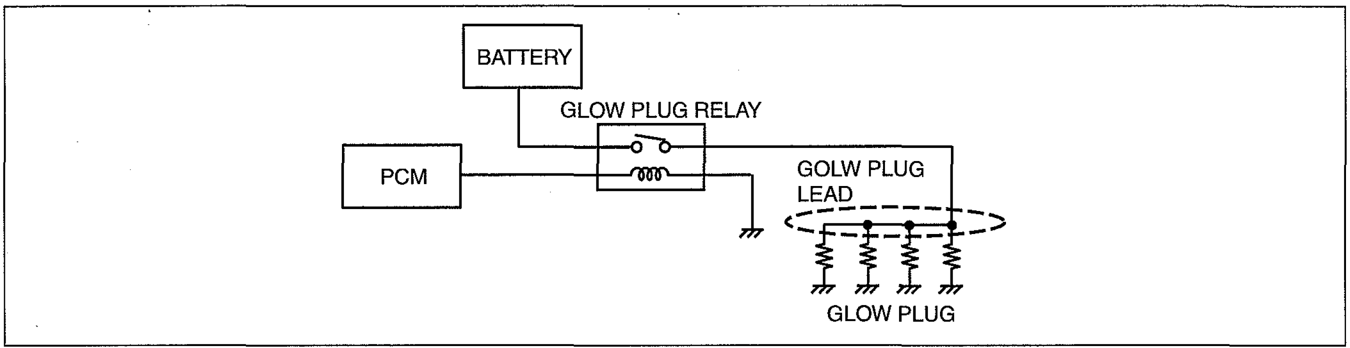
dcf011318601t08

- The glow system consists of the glow plug relay, glow plug lead, and glow plug.

GLOW SYSTEM OPERATION [WL-C, WE-C]

dcf011318601t09

- When the PCM energizes the glow plug relay, the glow plug activates, the circuit from the battery to the glow plug is established, and power is supplied to the glow plug via the glow plug lead. Due to this, the glow plug is heated and the combustion chamber is warmed up.



DGB113BTB005

GLOW PLUG FUNCTION [WL-C, WE-C]

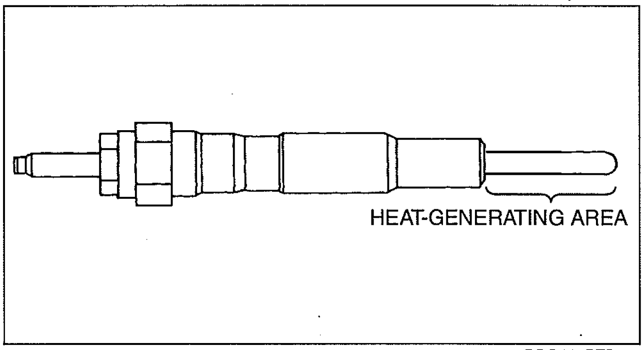
dcf011318601t10

- The glow plug warms up the air inside the combustion chamber by generating heat.

GLOW PLUG CONSTRUCTION [WL-C, WE-C]

dcf011318601t11

- Installed to the cylinder head.
- A stainless type glow plug has been adopted.

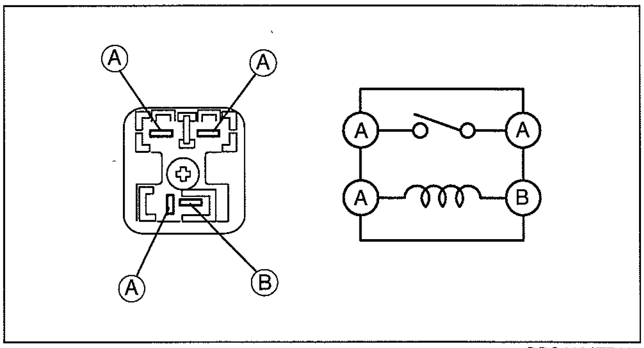


DBG113BTB804

GLOW PLUG RELAY OPERATION [WL-C, WE-C]

dcf011318601t12

- The glow plug relay activates according to the glow control signal from the PCM, and supplies/cuts the power to the glow plug.

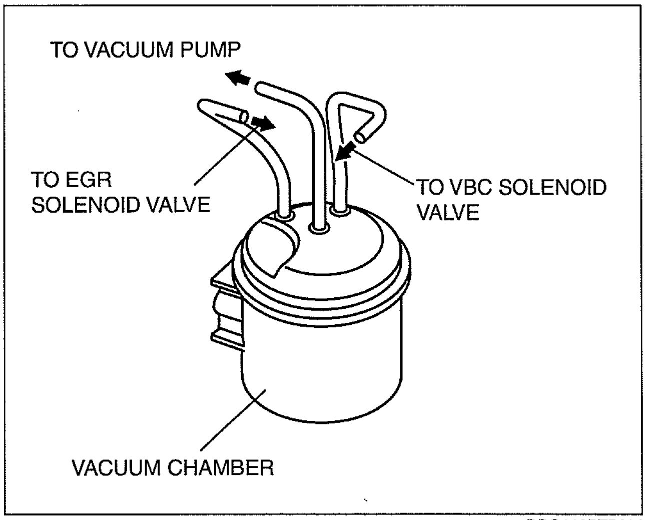


DBG113ATB105

VACUUM CHAMBER FUNCTION [WL-C, WE-C]

dcf011320330t01

- The vacuum chamber improves vacuum efficiency by reducing the pulsation generated when the air is drawn.



DBG113BTB302

VACUUM CHECK VALVE FUNCTION [WL-C, WE-C]

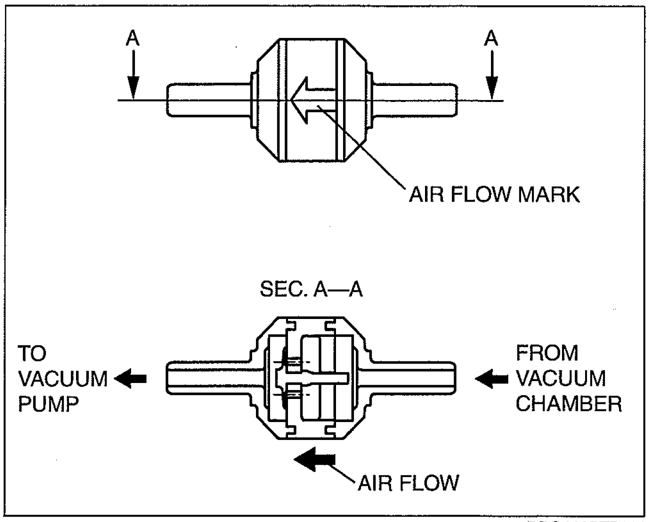
dcf011342910t01

- The one-way check valve prevents vacuum pressure inside the vacuum pump from flowing back to the vacuum chamber.

VACUUM CHECK VALVE CONSTRUCTION [WL-C, WE-C]

dcf011342910t02

- Mainly consists of the valve.



DBG113BTB301